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Understanding Wheat Lodging: Causes, Plant Failure Mechanisms and Management Approaches

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Introduction

Lodging in wheat is defined as the irreversible displacement of plant stems from their normal upright orientation, resulting in an inclined or horizontal position relative to the soil surface. This condition arises due to stem bending, stem breakage, or failure of the root system, most often triggered by strong winds combined with weak structural support. Lodging represents a major agronomic challenge as it substantially reduces grain yield, impairs grain quality, and ultimately lowers economic returns to farmers.

Wheat crops remain vulnerable to lodging from the booting stage until harvest, a sensitive window of nearly 60 days. Under severe conditions, plants may collapse entirely, forming a dense mat on the soil surface that complicates crop management and harvesting operations.

Importance of Wheat Production

Wheat (Triticum aestivum L.) is a cornerstone of global food security and serves as a primary energy source for a large proportion of the world's population. India is one of the leading wheat-producing nations, contributing over 15% of global wheat output and ranking second only to China. Average per capita wheat consumption in India ranges between 50 and 60 kg per year, highlighting its importance as a staple food after rice.

The Indo-Gangetic Plains Region (IGPR) is the principal wheat belt of India, extending from Rajasthan in the west to West Bengal in the east. The dominant rice–wheat cropping system in this region occupies approximately 10.5 million hectares. Lodging is a persistent problem in this intensive production system, particularly during the later stages of crop growth, and yield losses may range from 12% to as high as 80% under severe lodging conditions.

Causes of Lodging in Wheat

Several agronomic and environmental factors contribute to lodging in wheat. Early sowing exposes plants to prolonged low temperatures, which encourages excessive vegetative growth and increases susceptibility to bending under wind pressure. Similarly, the application of nitrogen fertilizers beyond recommended levels promotes lush growth with elongated and weaker stems, reducing mechanical strength.

Tall wheat cultivars, particularly those exceeding 120 cm in height, are inherently more prone to lodging, especially under high soil moisture conditions. Excessive irrigation and adverse weather events such as heavy rainfall, hailstorms, and strong winds during the dough stage further aggravate lodging risk.

Soil-related factors also play a significant role. Repeated intensive tillage can deteriorate soil structure, leading to compacted layers beneath loose topsoil. Such conditions restrict root penetration, weaken anchorage, and increase the likelihood of root lodging.

Mechanisms of Wheat Lodging

Lodging in wheat primarily occurs through two mechanical failure mechanisms associated with culm and root characteristics:

Stem lodging - This occurs when the bending moment exerted by wind and canopy weight exceeds the stem's mechanical strength, resulting in buckling or breakage, usually at the basal internodes.
Root lodging - This type arises when the root system fails to provide sufficient anchorage, particularly in water-saturated soils where root grip is reduced.

A specific form of stem lodging, known as necking, occurs when failure takes place just below the spike, causing the ear to droop while the lower stem remains upright.

Lodging Angle and Yield Reduction

The extent of yield loss due to lodging is closely linked to the angle formed between the stem and the vertical axis. Upright plants (0°) do not experience lodging-related losses. Minor yield reductions occur when lodging angles remain below 30°. At an angle of 45°, yield loss is approximately 25-50% of that observed when plants are almost completely flattened (80-90°). The greatest reduction in yield occurs when plants lie flat on the ground, severely restricting photosynthesis and grain development.

Adverse Effects of Lodging
Lodging disrupts canopy architecture, leading to reduced light interception and diminished photosynthetic efficiency. The altered microclimate within lodged crops promotes higher humidity, which favours pest infestations, disease development and pre-harvest sprouting.

Dry matter partitioning is also adversely affected, as assimilates are diverted away from developing grains. Grain quality deteriorates significantly, particularly when lodging occurs during early grain filling. Lodging reduces test weight, grain density, and the Hagberg Falling Number (HFN), an important parameter for assessing bread-making quality. An HFN value below 250 seconds is considered unsuitable for baking, and early lodging has been shown to cause substantial declines in this parameter.

Critical Growth Stages for Lodging
The impact of lodging varies with the growth stage at which it occurs. Lodging during early vegetative stages such as tillering or jointing may be partially corrected, as plants retain the ability to bend at nodes and regain an upright position. However, lodging at the booting stage or later is largely irreversible due to reduced cell elongation capacity.

The most severe yield losses occur when crops lodge during flowering or early grain filling, with reductions reaching up to 80%. Lodging near harvest generally causes smaller yield losses but still poses challenges for harvesting and grain quality maintenance.

Measurement of Lodging
Lodging is commonly assessed using lodging score and lodging index, which consider both the extent of area affected and the degree of inclination. Studies indicate that during the grain-filling period, approximately 0.5% of potential yield is lost for each 1% increase in lodged area per day, highlighting the cumulative impact of prolonged lodging.

Management Strategies for Wheat Lodging
Effective lodging management requires an integrated agronomic approach. Slightly delayed sowing can reduce excessive plant height and lodging risk, although extreme delays may reduce yield potential. Balanced fertilization is critical; nitrogen should be applied judiciously along with adequate phosphorus and potassium. Splitting nitrogen into three or four applications improves stem strength and reduces lodging susceptibility. Plant growth regulators (PGRs) such as chlormequat chloride, trinexapac-ethyl, ethephon and tebuconazole are effective when applied at the first node and flag leaf stages. These compounds restrict stem elongation,

enhance culm thickness, and improve root development. The adoption of semi-dwarf wheat varieties with stronger straw is one of the most reliable long-term solutions to lodging. Proper irrigation scheduling is equally important; excessive irrigation should be avoided, while timely light irrigation during the dough stage supports grain filling without increasing lodging risk. Conservation agriculture practices, including zero tillage and crop residue retention, improve soil structure, enhance root anchorage and further reduce lodging incidence.

Conclusion

Lodging remains a major constraint to sustainable wheat production, particularly in intensive cropping systems such as the rice-wheat system of the Indo-Gangetic Plains. It results from a complex interaction of varietal traits, nutrient management, irrigation practices, soil conditions, and climatic factors. Lodging significantly reduces both grain yield and quality, with the greatest damage occurring when it coincides with flowering or early grain filling. An integrated management strategy involving lodging-resistant varieties, balanced nutrient application, appropriate sowing time, regulated irrigation, use of plant growth regulators and conservation agriculture practices is essential for minimizing lodging and ensuring stable wheat productivity.

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